

1. Span Basics

- (a) Write the span $\left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \right\}$ as a linear combination of its basis vectors.

Answer:

$\text{span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \right\}$ contains any vector $\mathbf{u}$ that can be written as

$$\mathbf{u} = \alpha_1 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

Any vector whose last component is zero can be written in this form and any vector whose last component is nonzero cannot. Hence, the required span is the set of all vectors that can be written in the form $\begin{bmatrix} * \\ * \\ 0 \end{bmatrix}$.

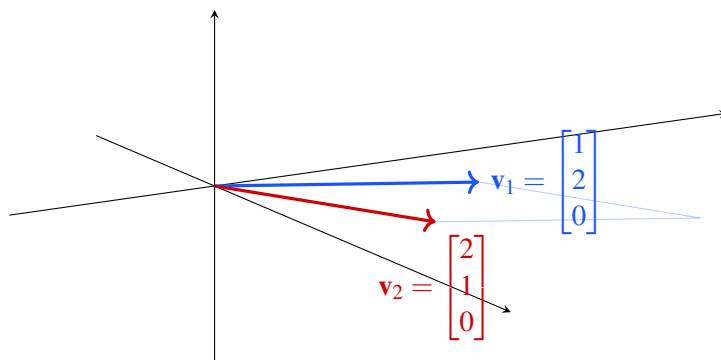


Figure 1: $\mathbf{v}_1$ and $\mathbf{v}_2$ span a 2D plane inside $\mathbb{R}^3$.

(b) Is $\begin{bmatrix} 5 \\ 5 \\ 0 \end{bmatrix}$ in $\text{span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} \right\}$?

Answer:

From the definition of span, we know that if we can express $\begin{bmatrix} 5 \\ 5 \\ 0 \end{bmatrix}$ as a linear combination of $\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$, it is in the span of those two vectors. Assume such a linear combination exists, then we can set up a vector equation

$$\alpha_1 \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} + \alpha_2 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \\ 0 \end{bmatrix}.$$

From this, we get the system of equations

$$\begin{cases} \alpha_1 + 2\alpha_2 = 5 \\ 2\alpha_1 + \alpha_2 = 5 \end{cases}$$

Solving this gives us $\alpha_1 = \frac{5}{3}$ and $\alpha_2 = \frac{5}{3}$, so we can conclude that the statement is true.

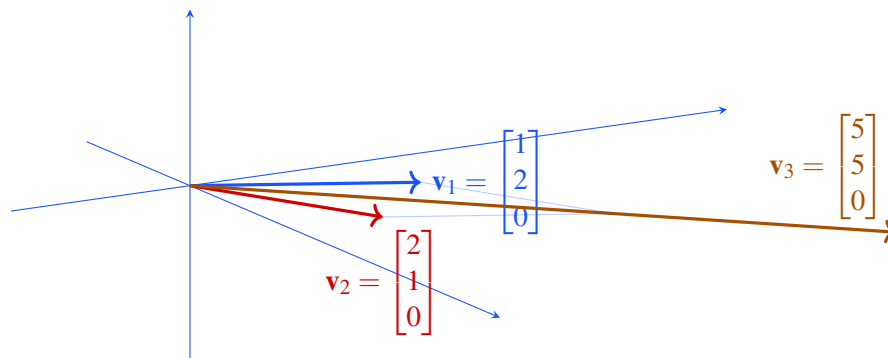


Figure 2: $\mathbf{v}_3$ is in the span of $\mathbf{v}_1$ and $\mathbf{v}_2$.

(c) What is a possible choice for $\mathbf{v}$ that would make $\text{span}\left\{\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \mathbf{v}\right\} = \mathbb{R}^3$?

Answer: From part (a), any vector whose last component is zero can be written as a linear combination of the two vectors $\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$. Hence if we include, for example, $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ into the set, then we should be able to reach any vector in $\mathbb{R}^3$. Any vector $\mathbf{v}$ whose last component is non-zero is a valid choice to achieve the desired span of $\mathbb{R}^3$.

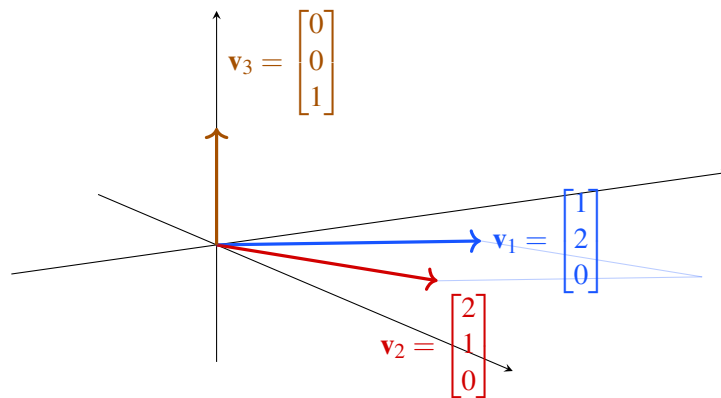


Figure 3: $\mathbf{v}_3$ is not in the span of $\mathbf{v}_1$ and $\mathbf{v}_2$, so adding it to the set allows us to reach any vector in $\mathbb{R}^3$.

2. Span Proof

Given some set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$, show the following:

$$\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} = \text{span}\{\alpha\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}, \text{ where } \alpha \text{ is a non-zero scalar}$$

In other words, we can scale our spanning vectors and not change their span.

Solution:

We defined the span to be:

$$\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} = \{\mathbf{q} \in \mathbb{R}^n \mid \mathbf{q} = \sum_{i=1}^n \alpha_i \mathbf{v}_i\}$$

for some scalars α_i .

Proving that an arbitrary vector in the span of the first set can be written as a linear combination of the vectors in the second set proves that the span of the first set is a subset of the span of the second set.

However, in order to prove equivalence, we need to have this in both directions—namely, we also need to prove that the span of the second set of vectors is a subset of the span of the first set.

Only when we have *both* of these conditions can we say that the two spans are equivalent.

Answer:

Suppose we have some arbitrary $\mathbf{q} \in \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$. For some scalars a_i :

$$\mathbf{q} = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_n\mathbf{v}_n = \left(\frac{a_1}{\alpha}\right)\alpha\mathbf{v}_1 + a_2\mathbf{v}_2 + \dots + a_n\mathbf{v}_n.$$

Scalar multiplication cancels out. Thus, we have shown that $\mathbf{q} \in \text{span}\{\alpha\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$. Therefore, we have $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} \subseteq \text{span}\{\alpha\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$. Now, we must show the other direction. Suppose we have some arbitrary $\mathbf{r} \in \text{span}\{\alpha\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$. For some scalars b_i :

$$\mathbf{r} = b_1(\alpha\mathbf{v}_1) + b_2\mathbf{v}_2 + \dots + b_n\mathbf{v}_n = (b_1\alpha)\mathbf{v}_1 + b_2\mathbf{v}_2 + \dots + b_n\mathbf{v}_n.$$

Thus, we have shown that $\mathbf{r} \in \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$. Therefore, we now have $\text{span}\{\alpha\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\} \subseteq \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$. Combining this with the earlier result, the spans are thus the same.

3. Exploring Dimension and Linear Independence

Let's consider the vector space $\mathbb{R}^k$ and a set of n vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ in $\mathbb{R}^k$.

- (a) For the first part of the problem, let $k > n$. Can $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ span the full $\mathbb{R}^k$ space? If so, justify it. If not, what conditions does it violate/what is missing?

Answer:

No, $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ cannot span the space $\mathbb{R}^k$. The dimension of $\mathbb{R}^k$ is k , so you would need k linearly independent vectors to describe the vector space. Since $n < k$, this is not possible.

Solution: (5 min)

- (b) Let $k = n$. Can $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ span the full $\mathbb{R}^k$ space? Why/why not? What conditions would we need?

Answer:

For the set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ to span $\mathbb{R}^k$ where $k = n$, the vectors must be linearly independent. This means that no vector in the set can be written as a linear combination of the others. If the vectors are linearly independent, they will form a basis for $\mathbb{R}^n$ and therefore span the full space.

If $n = k$ and the set $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is linearly independent, $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ can span the full $\mathbb{R}^k$ space.

Solution: (5 min)

Note to TAs: Emphasize here that *any* set of k linearly independent vectors would span $\mathbb{R}^k$. You could prompt a discussion on why this is true. Also tell students that they will prove the fact given in EECS16B.

- (c) Finally, let $k < n$. Can $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ span the full $\mathbb{R}^k$ space?

Answer:

The number of linearly independent vectors must be greater than or equal to the dimension of the space. If k out of the n vectors are linearly independent, then the set of vectors will be able to span the

full $\mathbb{R}^k$ space. However, if we have less than k linearly independent vectors in the set, then there is too much linear dependence for this set of vectors to be able to span the full $\mathbb{R}^k$ space.

Solution: (5 min)